

Changelog

1.5.0 (2023-11-13)

- Feature: Improve performance by using `spl_object_id()` on PHP 7.2+. (#267 by @samsonasik)
- Feature: Full PHP 8.3 compatibility. (#269 by @clue)
- Update tests for `ext-uv` on PHP 8+ and legacy PHP. (#270 by @clue and #268 by @SimonFrings)

1.4.0 (2023-05-05)

- Feature: Improve performance of `Loop` by avoiding unneeded method calls. (#266 by @clue)
- Feature: Support checking `EINTR` constant from `ext-pcntl` without `ext-sockets`. (#265 by @clue)
- Minor documentation improvements. (#254 by @nhedger)
- Improve test suite, run tests on PHP 8.2 and report failed assertions. (#258 by @WyriHaximus, #264 by @clue and #251, #261 and #262 by @SimonFrings)

1.3.0 (2022-03-17)

- Feature: Improve default `StreamSelectLoop` to report any warnings for invalid streams. (#245 by @clue)
- Feature: Improve performance of `StreamSelectLoop` when no timers are scheduled. (#246 by @clue)
- Fix: Fix periodic timer with zero interval for `ExtEvLoop` and legacy `ExtLibevLoop`. (#243 by @lucasnetau)
- Minor documentation improvements, update PHP version references. (#240, #248 and #250 by @SimonFrings, #241 by @dbu and #249 by @clue)
- Improve test suite and test against PHP 8.1. (#238 by @WyriHaximus and #242 by @clue)

1.2.0 (2021-07-11)

A major new feature release, see **release announcement**.

- Feature: Introduce new concept of default loop with the new `Loop` class. (#226 by @WyriHaximus, #229, #231 and #232 by @clue)

The `Loop` class exists as a convenient global accessor for the event loop. It provides all methods that exist on the `LoopInterface` as static methods and will automatically execute the loop at the end of the program:

```
$timer = Loop::addPeriodicTimer(0.1, function () {
    echo 'Tick' . PHP_EOL;
});

Loop::addTimer(1.0, function () use ($timer) {
    Loop::cancelTimer($timer);
    echo 'Done' . PHP_EOL;
});
```

The explicit loop instructions are still valid and may still be useful in some applications, especially for a transition period towards the more concise style. The `Loop::get()` method can be used to get the currently active event loop instance.

```
// deprecated
$loop = React\EventLoop\Factory::create();

// new
$loop = React\EventLoop\Loop::get();
```

- Minor documentation improvements and mark legacy extensions as deprecated. (#234 by @SimonFrings, #214 by @WyriHaximus and #233 and #235 by @nhedger)
- Improve test suite, use GitHub actions for continuous integration (CI), update PHPUnit config and run tests on PHP 8. (#212 and #215 by @SimonFrings and #230 by @clue)

1.1.1 (2020-01-01)

- Fix: Fix reporting connection refused errors with `ExtUvLoop` on Linux and `StreamSelectLoop` on Windows. (#207 and #208 by @clue)
- Fix: Fix unsupported `EventConfig` and `SEGFAULT` on shutdown with `ExtEventLoop` on Windows. (#205 by @clue)
- Fix: Prevent interval overflow for timers very far in the future with `ExtUvLoop`. (#196 by @PabloKowalczyk)
- Fix: Check `PCNTL` functions for signal support instead of `PCNTL` extension with `StreamSelectLoop`. (#195 by @clue)
- Add `.gitattributes` to exclude dev files from exports. (#201 by @reedy)
- Improve test suite to fix testing `ExtUvLoop` on Travis, fix Travis CI builds, do not install `libuv` on legacy PHP setups, fix failing test cases due to inaccurate timers, run tests on Windows via Travis CI and run tests on

PHP 7.4 and simplify test matrix and test setup. (#197 by @WyriHaximus and #202, #203, #204 and #209 by @clue)

1.1.0 (2019-02-07)

- New UV based event loop (ext-uv). (#112 by @WyriHaximus)
- Use high resolution timer on PHP 7.3+. (#182 by @clue)
- Improve PCNTL signals by using async signal dispatching if available. (#179 by @CharlotteDunois)
- Improve test suite and test suite set up. (#174 by @WyriHaximus, #181 by @clue)
- Fix PCNTL signals edge case. (#183 by @clue)

1.0.0 (2018-07-11)

- First stable LTS release, now following SemVer. We'd like to emphasize that this component is production ready and battle-tested. We plan to support all long-term support (LTS) releases for at least 24 months, so you have a rock-solid foundation to build on top of.

Contains no other changes, so it's actually fully compatible with the v0.5.3 release.

0.5.3 (2018-07-09)

- Improve performance by importing global functions. (#167 by @Ocrampus)
- Improve test suite by simplifying test bootstrap by using dev autoloader. (#169 by @lcobucci)
- Minor internal changes to improved backward compatibility with PHP 5.3. (#166 by @Donatello-za)

0.5.2 (2018-04-24)

- Feature: Improve memory consumption and runtime performance for `StreamSelectLoop` timers. (#164 by @clue)
- Improve test suite by removing I/O dependency at `StreamSelectLoopTest` to fix Mac OS X tests. (#161 by @nawarian)

0.5.1 (2018-04-09)

- Feature: New `ExtEvLoop` (PECL ext-ev) (#148 by @kaduev13)

0.5.0 (2018-04-05)

A major feature release with a significant documentation overhaul and long overdue API cleanup!

This update involves a number of BC breaks due to dropped support for deprecated functionality. We've tried hard to avoid BC breaks where possible and minimize impact otherwise. We expect that most consumers of this package will actually not be affected by any BC breaks, see below for more details.

We realize that the changes listed below may seem overwhelming, but we've tried to be very clear about any possible BC breaks. Don't worry: In fact, all ReactPHP components are already compatible and support both this new release as well as providing backwards compatibility with the last release.

- Feature / BC break: Add support for signal handling via new `LoopInterface::addSignal()` and `LoopInterface::removeSignal()` methods. (#104 by @WyriHaximus and #111 and #150 by @clue)

```
$loop->addSignal(SIGINT, function () {  
    echo 'CTRL-C';  
});
```

- Feature: Significant documentation updates for `LoopInterface` and `Factory`. (#100, #119, #126, #127, #159 and #160 by @clue, #113 by @WyriHaximus and #81 and #91 by @jsor)
- Feature: Add examples to ease getting started (#99, #100 and #125 by @clue, #59 by @WyriHaximus and #143 by @jsor)
- Feature: Documentation for advanced timer concepts, such as monotonic time source vs wall-clock time and high precision timers with millisecond accuracy or below. (#130 and #157 by @clue)
- Feature: Documentation for advanced stream concepts, such as edge-triggered event listeners and stream buffers and allow throwing `Exception` if stream resource is not supported. (#129 and #158 by @clue)
- Feature: Throw `BadMethodCallException` on manual loop creation when required extension isn't installed. (#153 by @WyriHaximus)
- Feature / BC break: First class support for legacy PHP 5.3 through PHP 7.2 and HHVM and remove all `callable` type hints for consistency reasons. (#141 and #151 by @clue)
- BC break: Documentation for timer API and clean up unneeded timer API. (#102 by @clue)

Remove `TimerInterface::cancel()`, use `LoopInterface::cancelTimer()` instead:

```
// old (method invoked on timer instance)
$timer->cancel();
```

```
// already supported before: invoke method on loop instance
$loop->cancelTimer($timer);
```

Remove unneeded `TimerInterface::setData()` and `TimerInterface::getData()`, use closure binding to add arbitrary data to timer instead:

```
// old (limited setData() and getData() only allows single variable)
$name = 'Tester';
$timer = $loop->addTimer(1.0, function ($timer) {
    echo 'Hello ' . $timer->getData() . PHP_EOL;
});
$timer->setData($name);
```

```
// already supported before: closure binding allows any number of variables
$name = 'Tester';
$loop->addTimer(1.0, function () use ($name) {
    echo 'Hello ' . $name . PHP_EOL;
});
```

Remove unneeded `TimerInterface::getLoop()`, use closure binding instead:

```
// old (getLoop() called on timer instance)
$loop->addTimer(0.1, function ($timer) {
    $timer->getLoop()->stop();
});
```

```
// already supported before: use closure binding as usual
$loop->addTimer(0.1, function () use ($loop) {
    $loop->stop();
});
```

- BC break: Remove unneeded `LoopInterface::isTimerActive()` and `TimerInterface::isActive()` to reduce API surface. (#133 by @clue)

```
// old (method on timer instance or on loop instance)
$timer->isActive();
$loop->isTimerActive($timer);
```

- BC break: Move `TimerInterface` one level up to `React\EventLoop\TimerInterface`. (#138 by @WyriHaximus)

```
// old (notice obsolete "Timer" namespace)
assert($timer instanceof React\EventLoop\Timer\TimerInterface);

// new
assert($timer instanceof React\EventLoop\TimerInterface);
```

- BC break: Remove unneeded `LoopInterface::nextTick()` (and internal `NextTickQueue`), use `LoopInterface::futureTick()` instead. (#30 by @clue)

```
// old (removed)
$loop->nextTick(function () {
    echo 'tick';
});
```

```
// already supported before
$loop->futureTick(function () {
    echo 'tick';
});
```

- BC break: Remove unneeded `$loop` argument for `LoopInterface::futureTick()` (and fix internal cyclic dependency). (#103 by @clue)

```
// old ($loop gets passed by default)
$loop->futureTick(function ($loop) {
    $loop->stop();
});
```

```
// already supported before: use closure binding as usual
$loop->futureTick(function () use ($loop) {
    $loop->stop();
});
```

- BC break: Remove unneeded `LoopInterface::tick()`. (#72 by @jsor)

```
// old (removed)
$loop->tick();
```

```
// suggested work around for testing purposes only
$loop->futureTick(function () use ($loop) {
    $loop->stop();
});
```

- BC break: Documentation for advanced stream API and clean up unneeded stream API. (#110 by @clue)

Remove unneeded `$loop` argument for `LoopInterface::addReadStream()` and `LoopInterface::addWriteStream()`, use closure binding instead:

```
// old ($loop gets passed by default)
$loop->addReadStream($stream, function ($stream, $loop) {
    $loop->removeReadStream($stream);
});
```

```
// already supported before: use closure binding as usual
$loop->addReadStream($stream, function ($stream) use ($loop) {
```

- ```

 $loop->removeReadStream($stream);
 });

```
- BC break: Remove unneeded `LoopInterface::removeStream()` method, use `LoopInterface::removeReadStream()` and `LoopInterface::removeWriteStream()` instead. (#118 by @clue)
 

```

// old
$loop->removeStream($stream);

// already supported before
$loop->removeReadStream($stream);
$loop->removeWriteStream($stream);

```
  - BC break: Rename `LibEventLoop` to `ExtLibeventLoop` and `LibEvLoop` to `ExtLibevLoop` for consistent naming for event loop implementations. (#128 by @clue)
  - BC break: Remove optional `EventBaseConfig` argument from `ExtEventLoop` and make its `FEATURE_FDS` enabled by default. (#156 by @WyriHaximus)
  - BC break: Mark all classes as final to discourage inheritance. (#131 by @clue)
  - Fix: Fix `ExtEventLoop` to keep track of stream resources (refcount) (#123 by @clue)
  - Fix: Ensure large timer interval does not overflow on 32bit systems (#132 by @clue)
  - Fix: Fix separately removing readable and writable side of stream when closing (#139 by @clue)
  - Fix: Properly clean up event watchers for `ext-event` and `ext-libev` (#149 by @clue)
  - Fix: Minor code cleanup and remove unneeded references (#145 by @seregazhuk)
  - Fix: Discourage outdated `ext-libevent` on PHP 7 (#62 by @cboden)
  - Improve test suite by adding forward compatibility with PHPUnit 6 and PHPUnit 5, lock Travis distro so new defaults will not break the build, improve test suite to be less fragile and increase test timeouts, test against PHP 7.2 and reduce `fwrite()` call length to one chunk. (#106 and #144 by @clue, #120 and #124 by @carusogabriel, #147 by nawarian and #92 by @kelunik)
  - A number of changes were originally planned for this release but have been backported to the last v0.4.3 already: #74, #76, #79, #81 (refs #65, #66, #67), #88 and #93

### 0.4.3 (2017-04-27)

- Bug fix: Bugfix in the usage sample code #57 (@dandelionred)
- Improvement: Remove branch-alias definition #53 (@WyrHaximus)
- Improvement: StreamSelectLoop: Use fresh time so Timers added during stream events are accurate #51 (@andrewminer)
- Improvement: Avoid deprecation warnings in test suite due to deprecation of getMock() in PHPUnit #68 (@martinschroeder)
- Improvement: Add PHPUnit 4.8 to require-dev #69 (@shaunbramley)
- Improvement: Increase test timeouts for HHVM and unify timeout handling #70 (@clue)
- Improvement: Travis improvements (backported from #74) #75 (@clue)
- Improvement: Test suite now uses socket pairs instead of memory streams #66 (@martinschroeder)
- Improvement: StreamSelectLoop: Test suite uses signal constant names in data provider #67 (@martinschroeder)
- Improvement: ExtEventLoop: No longer suppress all errors #65 (@mami-ciek)
- Improvement: Readme cleanup #89 (@jsor)
- Improvement: Restructure and improve README #90 (@jsor)
- Bug fix: StreamSelectLoop: Fix erroneous zero-time sleep (backport to 0.4) #94 (@jsor)

### 0.4.2 (2016-03-07)

- Bug fix: No longer error when signals sent to StreamSelectLoop
- Support HHVM and PHP7 (@ondrejmirtes, @cebe)
- Feature: Added support for EventConfig for ExtEventLoop (@steverhoades)
- Bug fix: Fixed an issue loading loop extension libs via autoloader (@czarpino)

### 0.4.1 (2014-04-13)

- Bug fix: null timeout in StreamSelectLoop causing 100% CPU usage (@clue)
- Bug fix: v0.3.4 changes merged for v0.4.1

### 0.4.0 (2014-02-02)

- Feature: Added EventLoopInterface::nextTick(), implemented in all event loops (@jmallo)
- Feature: Added EventLoopInterface::futureTick(), implemented in all event loops (@jmallo)
- Feature: Added ExtEventLoop implementation using pecl/event (@jmallo)

- BC break: Bump minimum PHP version to PHP 5.4, remove 5.3 specific hacks
- BC break: New method: `EventLoopInterface::nextTick()`
- BC break: New method: `EventLoopInterface::futureTick()`
- Dependency: Autoloading and filesystem structure now PSR-4 instead of PSR-0

### 0.3.5 (2016-12-28)

This is a compatibility release that eases upgrading to the v0.4 release branch. You should consider upgrading to the v0.4 release branch.

- Feature: Cap min timer interval at 1µs, thus improving compatibility with v0.4 (#47 by @clue)

### 0.3.4 (2014-03-30)

- Bug fix: Changed StreamSelectLoop to use non-blocking behavior on tick() (@astephens25)

### 0.3.3 (2013-07-08)

- Bug fix: No error on removing non-existent streams (@clue)
- Bug fix: Do not silently remove feof listeners in LibEvLoop

### 0.3.0 (2013-04-14)

- BC break: New timers API (@nrk)
- BC break: Remove check on return value from stream callbacks (@nrk)

### 0.2.7 (2013-01-05)

- Bug fix: Fix libevent timers with PHP 5.3
- Bug fix: Fix libevent timer cancellation (@nrk)

### 0.2.6 (2012-12-26)

- Bug fix: Plug memory issue in libevent timers (@cameronjacobson)
- Bug fix: Correctly pause LibEvLoop on stop()

### 0.2.3 (2012-11-14)

- Feature: LibEvLoop, integration of `php-libev`

### 0.2.0 (2012-09-10)

- Version bump

### **0.1.1 (2012-07-12)**

- Version bump

### **0.1.0 (2012-07-11)**

- First tagged release